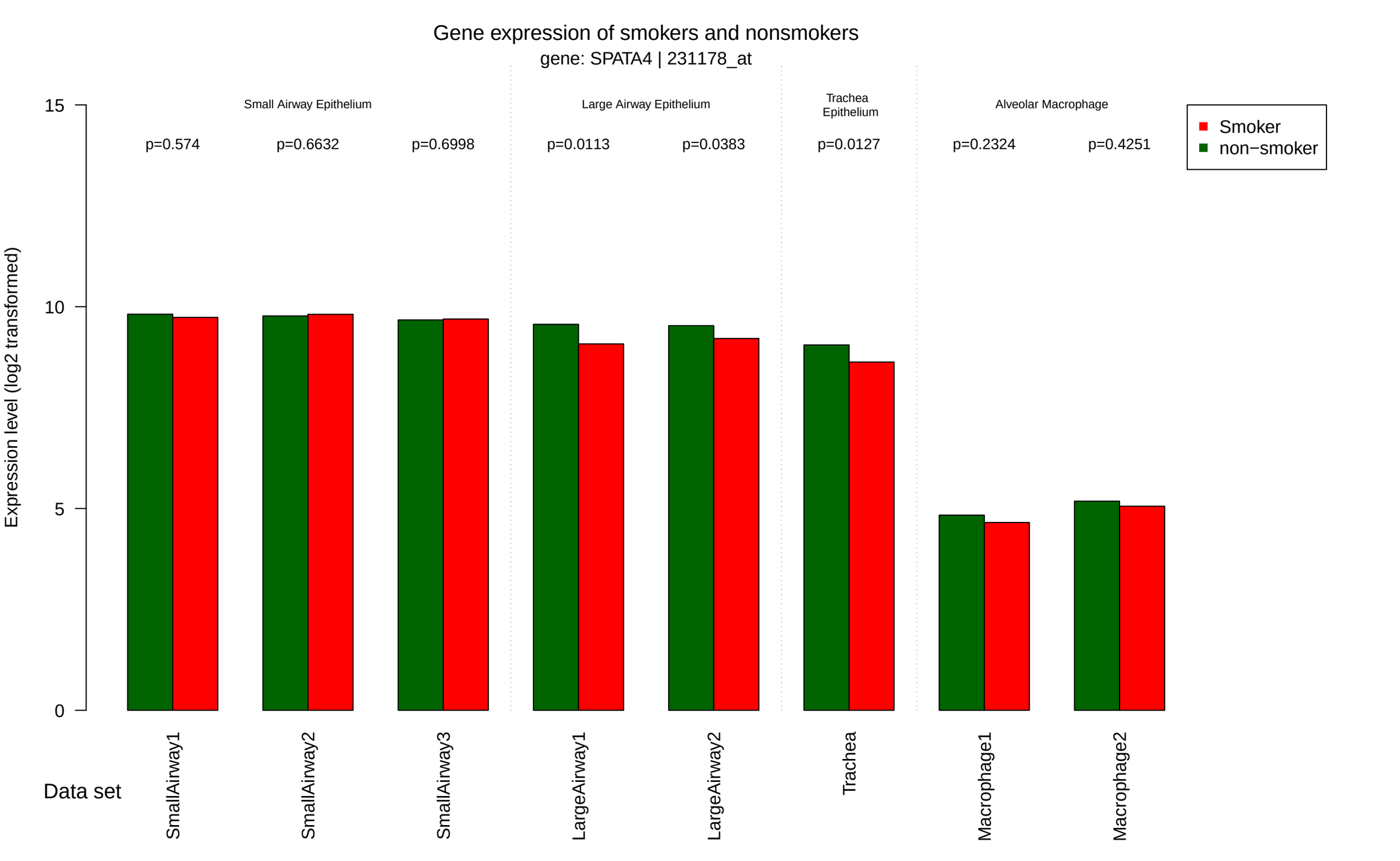
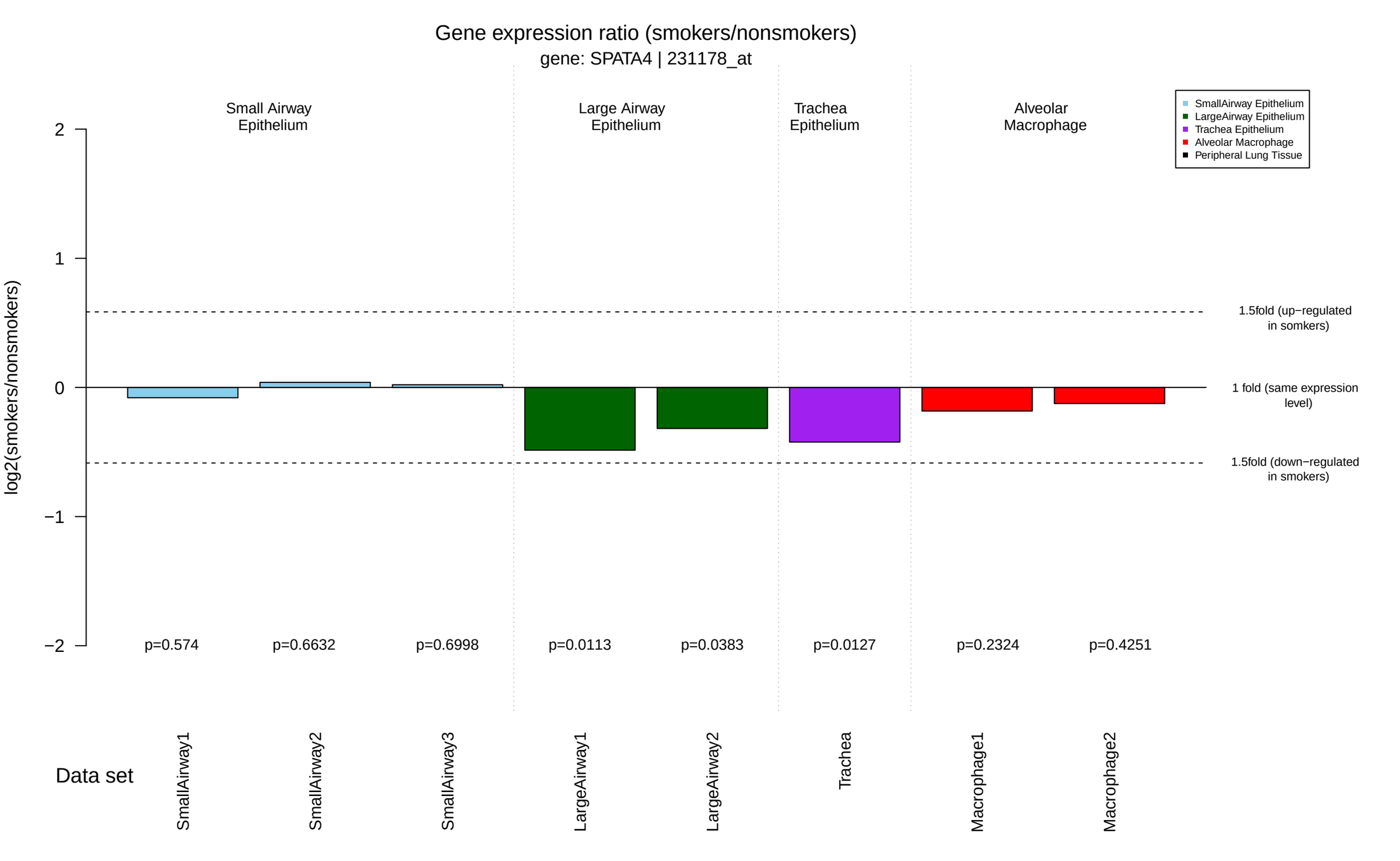
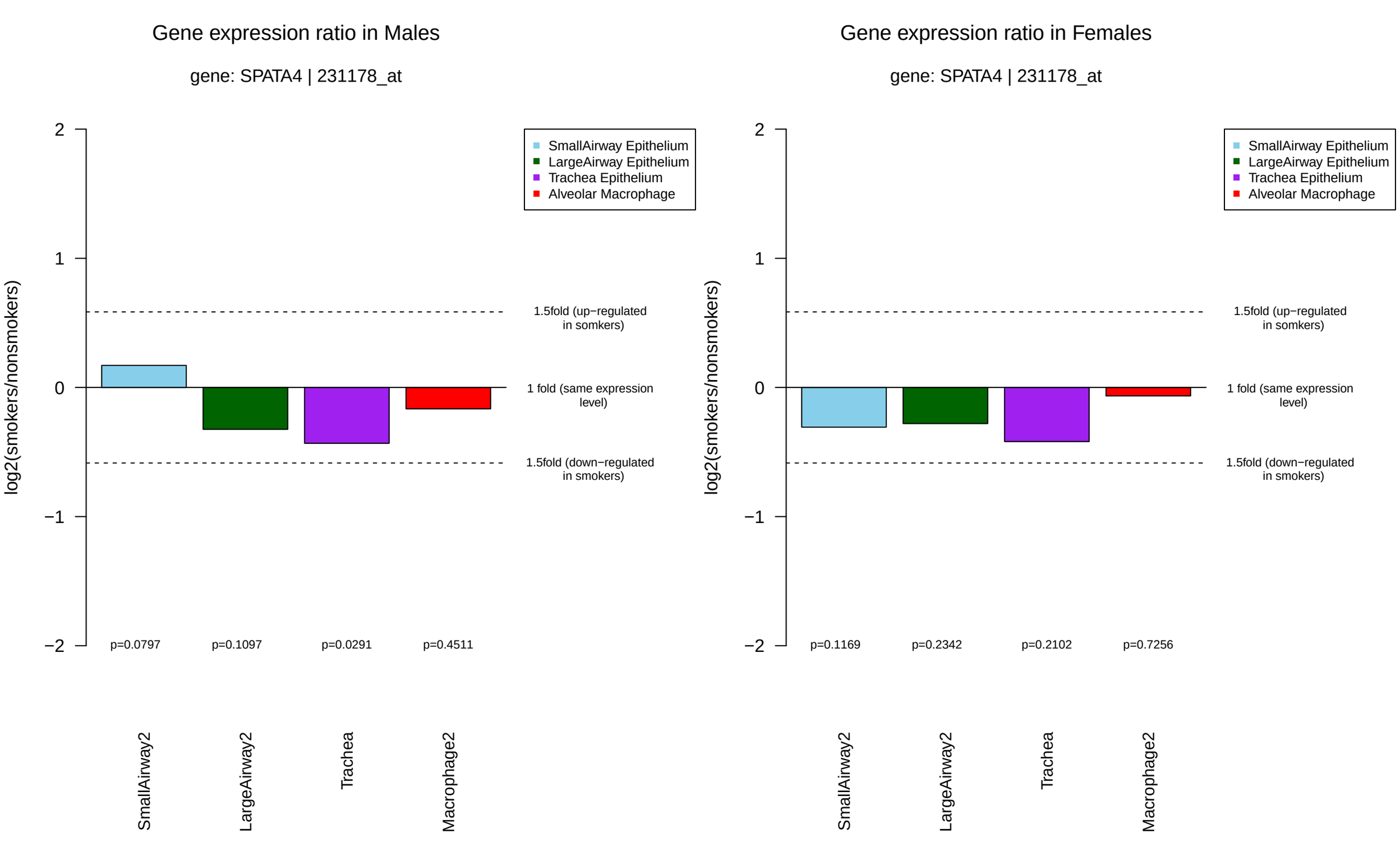
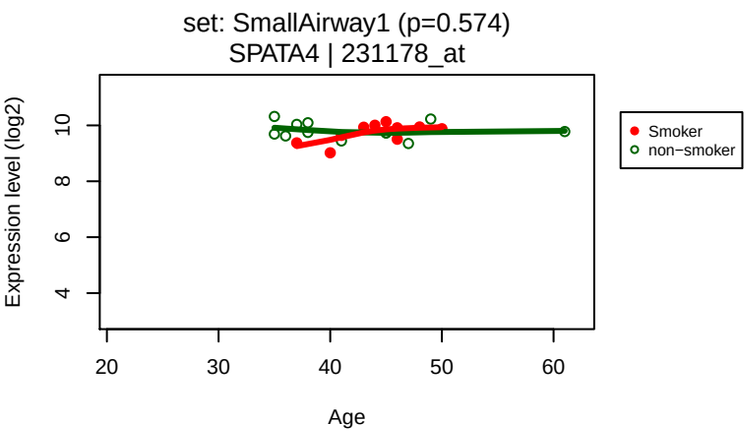
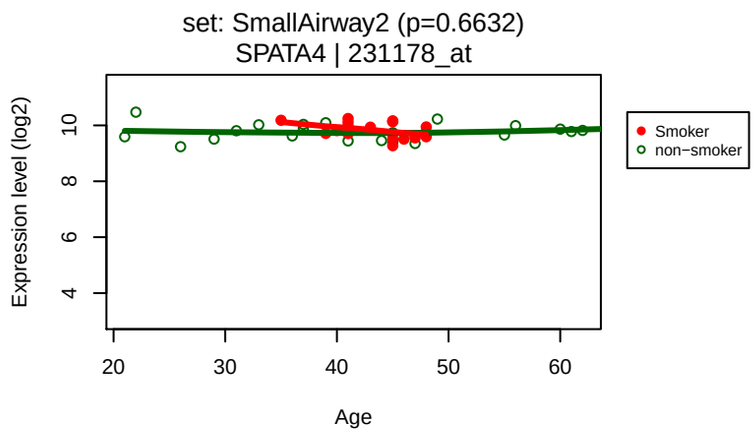
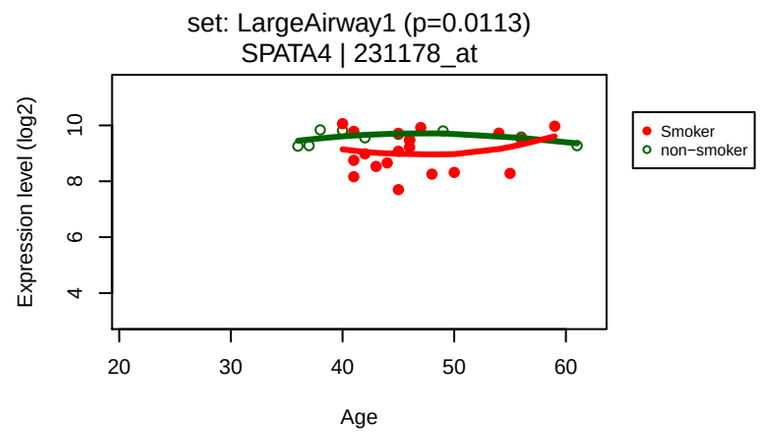

Gene expression of smokers and nonsmokers
gene: SPATA4 | 231178_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.574
p=0.6632
p=0.6998
p=0.0113
p=0.0383
p=0.0127
p=0.2324
p=0.4251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

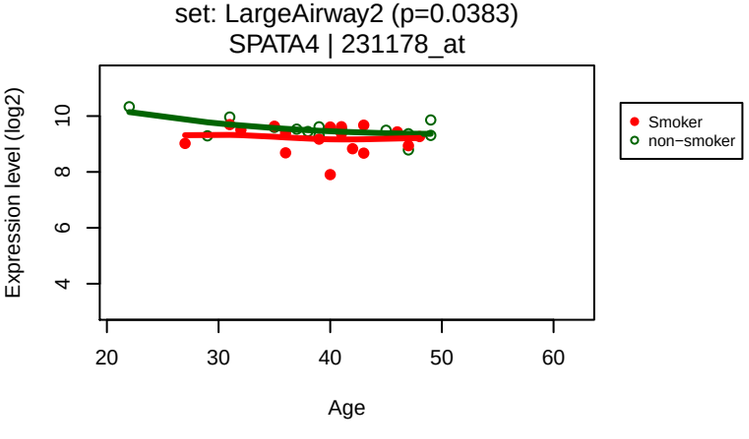

Gene expression ratio (smokers/nonsmokers)
gene: SPATA4 | 231178_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.574
p=0.6632
p=0.6998
p=0.0113
p=0.0383
p=0.0127
p=0.2324
p=0.4251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

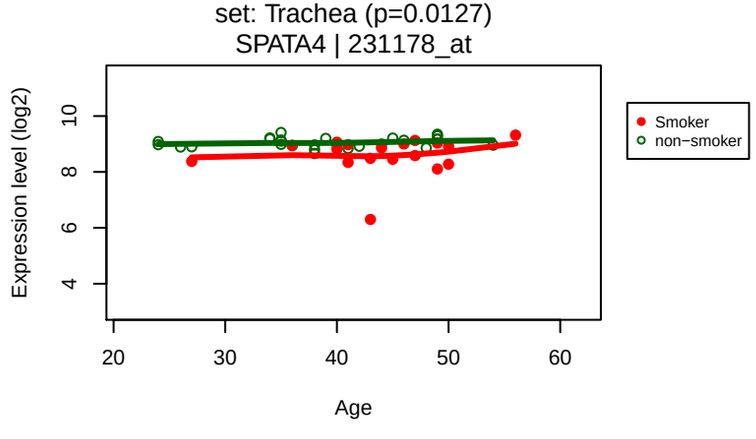

Gene expression ratio in Males
gene: SPATA4 | 231178_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0797
p=0.1097
p=0.0291
p=0.4511
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA4 | 231178_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1169
p=0.2342
p=0.2102
p=0.7256
SmallAirway2
LargeAirway2
Trachea
Macrophage2

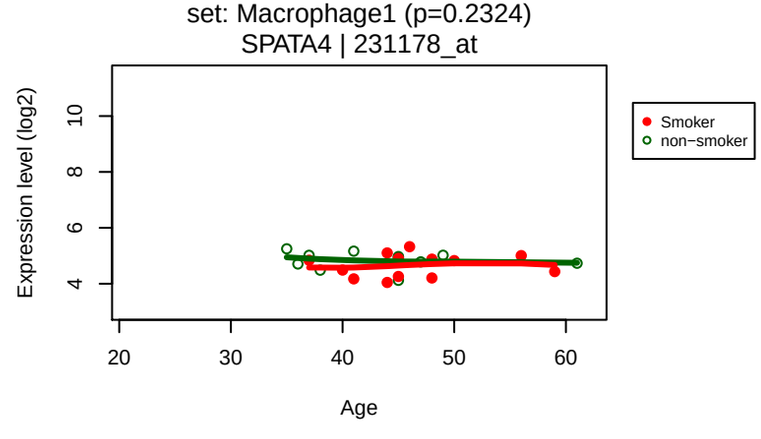

set: SmallAirway1 (p=0.574)
SPATA4 | 231178_at
Expression level (log2)
Age
Smoker
non−smoker

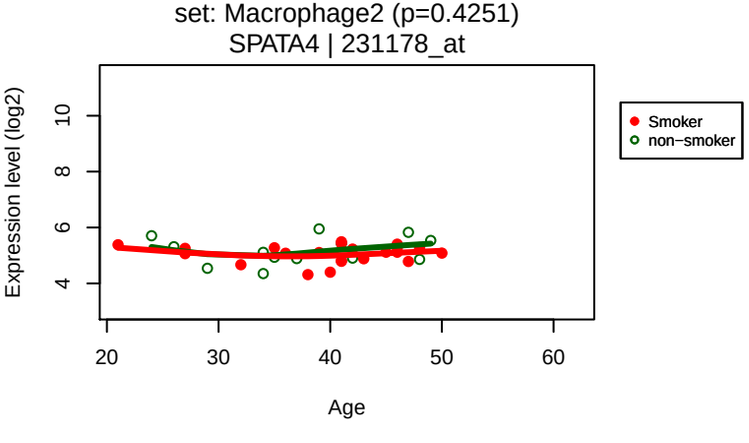

set: SmallAirway2 (p=0.6632)
SPATA4 | 231178_at
Expression level (log2)
Age
Smoker
non−smoker

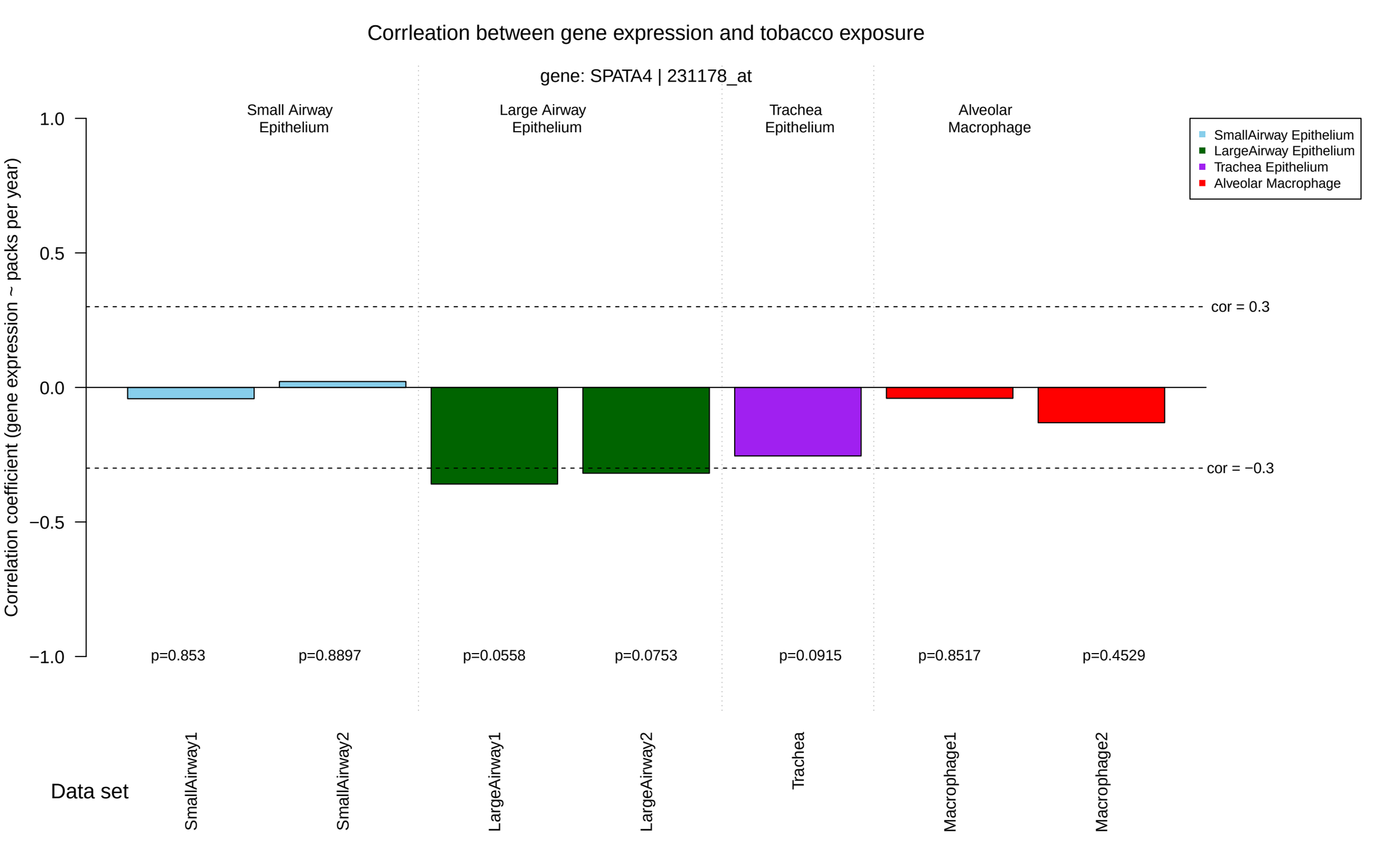

set: LargeAirway1 (p=0.0113)
SPATA4 | 231178_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0383)
SPATA4 | 231178_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0127)
SPATA4 | 231178_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2324)
SPATA4 | 231178_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4251)
SPATA4 | 231178_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA4 | 231178_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.853
p=0.8897
p=0.0558
p=0.0753
p=0.0915
p=0.8517
p=0.4529
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2